

Soln Su24 sec 3/4 quiz1 set B

① a) Total = 4.5×10^6

$$X = 4.5 \times 10^6 \times \frac{15}{100}$$

$$Z = 4.5 \times 10^6 \times \frac{40}{100}$$

$$Y = 4.5 \times 10^6 \times \frac{15}{100}$$

$$W = 4.5 \times 10^6 \times \frac{30}{100}$$

$$\begin{aligned} \text{Total time (AMD)} &= 6.75 \times 10^5 \times 1 + 6.75 \times 10^5 \times 2 \\ &\quad + 1.8 \times 10^6 \times 1 + 1.35 \times 10^6 \times 4 \\ &\quad \hline &2 \times 10^9 \end{aligned}$$

$$= 4.6125 \times 10^{-3}$$

$$\begin{aligned} \text{Total time (RVZ)} &= 6.75 \times 10^5 \times 2 + 6.75 \times 10^5 \times 1 \\ &\quad + 1.8 \times 10^6 \times 3 + 1.35 \times 10^6 \times 2 \\ &\quad \hline &2 \times 10^9 \\ &= 5.0625 \times 10^{-3} \end{aligned}$$

$$\text{CPI (AMD)} = \frac{4.6125 \times 10^{-3} \times 2 \times 10^9}{4.5 \times 10^6} = 2.05$$

$$\text{CPI (RVZ)} = \frac{5.0625 \times 10^{-3} \times 2 \times 10^9}{4.5 \times 10^6} = 2.25$$

$$\textcircled{b} \text{ Clock cycle (AMD)} = 6.75 \times 10^5 \times 1 \\ + 6.75 \times 10^5 \times 2 \\ + 1.8 \times 10^6 \times 1 \\ + 1.35 \times 10^6 \times 4 \\ = 9.225 \times 10^6$$

$$\text{Clock cycle (RYZ)} = 6.75 \times 10^5 \times 2 \\ + 6.75 \times 10^5 \times 1 \\ + 1.8 \times 10^6 \times 3 \\ + 1.35 \times 10^6 \times 2 \\ = 1.0125 \times 10^7$$

$$\textcircled{c} \text{ Exec (AMD)} = 4.6125 \times 10^{-3} \text{ sec} \\ \text{Exec (RYZ)} = 5.0625 \times 10^{-3} \text{ sec}$$

$$\text{AMD Spec Ratio} = \sqrt{\frac{18700}{4.6125 \times 10^{-3}}}$$

$$\text{RYZ Spec Ratio} = \sqrt{\frac{18700}{5.0625 \times 10^{-3}}}$$

Due to this being a ratio, it won't matter whether we use reference execution or execution reference.

② diameter = 16 cm radius = 8 cm

$$\text{die area} = \frac{\text{wafer area}}{\text{dies per wafer}}$$

$$= \frac{\pi \times 8^2}{84} = 2.3935 \text{ cm}^2$$

$$\text{Yield} = \frac{1}{\left(1 + \left(0.050 \times \frac{2.3935}{2}\right)\right)^2}$$
$$= 0.89026$$

$$\text{Cost per die} = \frac{24}{84 \times 0.890} = 0.320 \quad (\text{Ans})$$

③ a) Algorithm, Programming language, Compiler, ISA/Processor & memory system, I/O system.

b) Datapath, Control, Cache memory

c) Elapsed time: Total response time, including all aspects (I/O, OS overhead, idle time)

Execution time: Only time spent processing given job.

d) i) reduce number of clock cycles

ii) Increase clock rate

iii) Trade off clock rate against cycle count

e) Every 2 years, # of transistors double, $\frac{\text{size}}{2}$, computing power doubles